



Hands-on Principal Engineer & Architect · 19+ yrs · Python + Node.js · Governing AI-Assisted Delivery (Claude Code · Codex) · Agentic AI Engineering (LangChain · LangGraph · MCP · Evaluations & Guardrails) · Cloud-Native AWS · Commerce & Multi-Tenant SaaS

SUMMARY

Hands-on Principal Engineer and Architect with **19+ years** of end-to-end ownership — taking an ambiguous problem, defining the architecture across a whole domain, and shipping the first cut myself. Expert-fluent in **Python** (Django, FastAPI, LangChain, LangGraph) and **Node.js**, with Go and Java also in production. Most recently defined and governed an **Artificial Intelligence (AI)-assisted development practice** across **20 engineers** at VoltusWave — Claude Code and OpenAI Codex for architectural exploration, first-cut implementations and test scaffolding, all under hands-on review, on top of a **Continuous Integration / Continuous Delivery (CI/CD)** pipeline I modernised at Deque. Agentic AI is the day job: a **10-query decision-intelligence platform** in Python orchestrated across **4 reusable archetypes** on Amazon Neptune, dense-vector precedent retrieval under measured evaluation, narrative guardrails built on a pinned model and a field-level claim grammar, and real **Model Context Protocol (MCP)** servers for **Large Language Model (LLM)** tool use. Cloud-native on **Amazon Web Services (AWS)** at depth — Elastic Container Service Fargate, Kinesis, DynamoDB, Neptune, OpenSearch, Kubernetes, Terraform — with Datadog observability, service-level objectives and on-call rotation, and **Health Insurance Portability and Accountability Act (HIPAA)** plus **International Organization for Standardization (ISO) 27001** security. Built engineering **0 → 20+** across full stack, mobile, quality assurance (QA), and DevOps; **5+ years at Deque Software** leading distributed **United States–India** teams on multi-tenant **Software as a Service (SaaS)** serving **20+ enterprise clients**; earlier, travel-commerce work that lifted **hotel booking conversions 65%**.

Key Skills

Technology Leadership & Strategy

- Technology Strategy & Architecture
- Engineering Team Building & Organisational Scaling
- **Cross-Domain Solution Architecture — Defining & Delivering Technology Solutions End to End**
- **Governing AI-Assisted Development Across an Engineering Organisation**
- **Product & Platform Strategy Across Commerce, Subscription & Multi-Tenant SaaS**
- Cross-Functional Leadership Across Geographies & Time Zones (Product, User Experience (UX), Analytics)

Engineering Excellence

- Multi-Tenant SaaS Architecture
- System Design at Scale (Distributed Systems, Relational Databases, Application Programming Interfaces (APIs))
- Cross-Team Technical Leadership (6 sub-teams, company-wide standards)
- **AI Code Generation in the Delivery Workflow (Claude Code, OpenAI Codex — Under Hands-On Review)**
- **Agentic AI Engineering (Orchestration on LangChain / LangGraph, Large Language Model (LLM) Tool Use, Guardrails)**
- Expert Fluency in Python (Django, FastAPI, LangChain, LangGraph) and Node.js
- **Model Context Protocol (MCP) Servers & Tool-Use Integration**
- Security Paradigms (HIPAA, ISO 27001, Role-Based Access Control, Least-Privilege)
- **AI Evaluation Frameworks & Guardrail Design (Retrieval Evaluation, Field-Level Claim Grammar, Replay Determinism)**
- Shared Platform APIs (Contracts, Backward Compatibility)
- **Cloud-Native AWS (Elastic Container Service Fargate, Kinesis, DynamoDB, Relational Database Service, OpenSearch, Neptune)**
- **Go and Java Production Services (Node.js → Go Rewrite Sustaining 100,000 Concurrent Users)**
- **Deployment Infrastructure (Kubernetes, Terraform, Deploy Under 10 Minutes)**
- **Observability & Resilience (Datadog, Service-Level Objectives, Incident Response, On-Call Rotation)**

Business & Delivery

- Agile / Scrum Delivery Management
- **Commerce & Subscription Domain Delivery (Travel Booking Funnel, 25-Supplier Integration, Enterprise SaaS Tenants)**
- **Performance Optimization & Cost Management (Infrastructure Cost Down 25–30%)**

PROFESSIONAL EXPERIENCE

Principal Software Architect

Mar '25 - Apr '26

Voltuswave Technologies India Pvt Ltd

Hyderabad

- Embedded a governed **Claude Code + OpenAI Codex** practice across **20 engineers** — architectural exploration, first-cut implementations, and test scaffolding under hands-on review
- Pioneered **Context Graph** on Amazon Neptune with graph neural network models — virtual care decision-intelligence across **80+ chronic conditions** for **10,000+ patients**
- Grew engineering **0 → 20+** across full stack, mobile, quality assurance (QA), AI, and DevOps — **6 sub-teams** reporting through tech leads
- Architected Chat Service on **Amazon Web Services Kinesis** with two-stream Enhanced Fan-Out (**4 → 1,200 shards**) on Elastic Container Service Fargate at 70% Spot

- Scaled Chat Service from **500 to 100,000 concurrent users** at ~27 ms avg latency, P95 under **16 ms**, 100% success — k6-validated
- Spearheaded the complete rewrite of the production **Node.js** Chat Service into **Go** for **100K-scale** concurrency
- Reimagined data architecture across **DynamoDB, PostgreSQL, OpenSearch** with optimised partition-key + indexing design — **P95 < 30 ms** at 100K-user load
- Drove cross-domain technical architecture across **6 sub-teams** — company-wide adoption of **Go and DynamoDB** with design-doc discipline holding one coherent ecosystem
- Owned end-to-end **React** web delivery and the **React Native** client shipped to **2 app stores** with Firebase Cloud Messaging and Apple Push Notification service
- Operationalised **Amazon Web Services** infrastructure as code — Kubernetes, Terraform, Elastic Container Service Fargate at 70% Spot — deploy under **10 minutes**, cost down **25–30%**
- Improved production reliability across **6 sub-teams** — **Service-Level Objective (SLO)** targets, **Datadog** observability, incident response, and on-call rotation hygiene
- Established **Health Insurance Portability and Accountability Act (HIPAA)** framework across **10 production pipelines** — data protection, access governance, audit trails for clinical workflows
- Launched a **10-query decision-intelligence platform** in **Python (FastAPI, LangChain, LangGraph)** — orchestration across **4 reusable archetypes** on Amazon Neptune
- Instituted event-sourced architecture (**DynamoDB single-table**) with deterministic parquet snapshot exporter and correction-event backbone — **P95 ≤ 5 min** targeted rebuild
- Secured **multi-tenant SaaS isolation** across **10 production pipelines** — least-privilege access, TraversalSource wrapper, runtime assertion, per-tenant Neptune for healthcare data
- Deployed **large language model narrative guardrails** — pinned model + temperature=0 + chain hash + field-level claim grammar — **100% uncited claims rejected**
- Optimized precedent retrieval on **Elasticsearch dense-vector search** with self-hosted SageMaker embedders — warm-path **P99 < 200 ms**, **Normalized Discounted Cumulative Gain (NDCG)@10 ≥ 0.75**
- Strengthened **statistical rigor** across **6 production pipelines** — propensity matching on a shared match_manifest_id, **B=2000** bootstrap resamples, **Rosenbaum γ + E-value** sensitivity battery

Lead SDE/Sr. Staff Engineer / Staff Software Engineer Dec '19 - Feb '25

Deque Software Hyderabad

- Led distributed **United States–India** teams across engineering, QA, and support on **multi-tenant SaaS**, cutting time-to-market **2+ weeks** per sprint
- Designed shared platform APIs on a **multi-tenant SaaS** — Jira integration and enterprise contracts secured **20+ enterprise clients** with strict backward-compatibility discipline
- Migrated PostgreSQL to **Amazon Web Services (AWS) Relational Database Service (RDS)**, cutting backups **3-4 hrs/day** and deploy errors **40%**
- Modernised **CI/CD** from Jenkins to GitHub Actions — earning **ISO 27001** and **50% faster releases**
- Engineered an end-to-end automated testing framework in **Node.js** (Puppeteer), **reducing regression testing time by over 50%** across every release
- Raised API quality across **20+ enterprise-client SaaS** tenants — instilling a code-review standard, documentation discipline, and scalable contracts across engineering and product

Software Engineer III Aug '18 - Jul '19

Rocket Software Bangalore

- Supervised a **5-engineer team** for a **United States-headquartered** product company — performance appraisals, delivery planning, and code-review mentoring across time zones
- Reworked IBM Configuration Manager for z/Systems and Linux/Unix/Windows (LUW) into **Java** microservices with **Representational State Transfer (REST)** APIs, **cutting reported bugs over 50%**
- Increased product adoption by **over 20%** through RESTful APIs and microservices architecture
- Expanded platform support to Windows and Linux distributions (Ubuntu, CentOS, Red Hat Enterprise Linux (RHEL)), **driving over 20% sales uplift**

Co-Founder and VP of Technology Mar '17 - Apr '18

Voltuswave Technologies India Pvt Ltd Hyderabad

- Co-founded VoltusWave and architected the first application Platform as a Service (**aPaaS**) product end to end across **2 platforms** — web and Android
- Built and scaled engineering team from scratch to **35 members**, mentoring freshers and seniors while establishing delivery processes
- Delivered end-to-end development, deployment, and operations on AWS Elastic Container Service — highly scalable solutions that won **8 early enterprise clients**

Senior Software Engineer Mar '16 - Mar '17

Teletext India Private Limited, Hyderabad Hyderabad

- Orchestrated migration of **travel-commerce supplier integration** from legacy .NET 4.0 to **MuleSoft Enterprise Service Bus (ESB)** across **25 suppliers**
- Shipped a hotel content enrichment pipeline that lifted **hotel booking conversions 65%** across the online travel **commerce** funnel
- Retired a third-party product for an in-house **Elasticsearch / Logstash / Kibana (ELK)** observability stack, cutting operational costs **46%**

Technical Architect May '15 - Nov '15

CURA Software Solutions Hyderabad

- Introduced **Linux/Unix deployment** for the Governance, Risk & Compliance platform, cutting running costs and winning **12 new customers within 2 months**
- Ported the data model from **Microsoft SQL Server** to **MySQL**, enabling the low-cost Linux deployment

Technical Lead Jul '13 - May '15

innRoad India Hotel Software Pvt. Ltd. Hyderabad

- Converted a **legacy ASP.NET 2.0** application into a modern **single-page application** using Knockout.js and a Boilerplate JavaScript framework
- Productized **REST-based WCF** services that enabled and monetised customer integrations
- Codified engineering standards and mentored the team through a full **product-modernisation effort**

Software Architect	Jun '12 - May '13
McDonald's Corporation	OakBrook, IL
- Modeled the schema and governance framework for an enterprise Master Data Management (MDM) program - Forged interactive user-interface screens with drill-down, dynamic drop-downs, and full row-level editing - Partnered with data owners to deliver a robust enterprise data-governance system across the organisation	

Senior .NET Developer	Apr '08 - May '12
El Paso Corporation	Houston, TX
- Rebuilt a legacy Citrix-hosted Java application into a modern .NET 2.0 stack on SQL Server 2005 - Steered a scalable, maintainable .NET platform that replaced the outdated system while preserving core business functions - Anchored large-scale enterprise modernisation for a high-stakes energy-sector client	

.NET Developer	Apr '07 - Apr '08
LyntonWeb	Houston, TX
Developed Paymaster, a PayPal-style payments application, for a Mexican financial institution	

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning	Jan '21
IIIT Bangalore	
Master's in Computer Science	Aug '04 - Dec '07
University of Houston	Houston, Texas

CERTIFICATION

PG Diploma in Artificial Intelligence and Machine Learning, IIIT Bangalore	
--	--